PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

School of Information Science

Department of Master of Computer Applications

Smart Community Health Monitoring and Early Warning System For Water-Borne Diseases in Rural Northeast India

PROJECT REPORT

Submitted By

NIKHITHA V SHETTY - 20242MCA0137

Under the guidance of

Dr N Baskar

ASSOCIATE PROFESSOR OF SOIS

MASTER OF COMPUTER APPLICATIONS

SCHOOL OF INFORMATION SCIENCE

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BONAFIDE CERTIFICATE

Certified that this project titled **" Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India"** is a bonafide work of "NIKHITHA V SHETTY (20242MCA0137)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications** during 2025-26.

Dr N Baskar	**Dr. M Renuka Devi**
Project Guide	HOD
Master of Computer Applications	Master of Computer Applications
Presidency University	Presidency University

Dr. R Mahalakshmi

Associate Dean

School of Information science

Presidency Universit

Examiners

Sl. no.	Name	Signature	Date
1	Dr. M. Anand Kumar		
2			

DECLARATION

I'm the students of second year Master of Computer Applications at Presidency University, Bengaluru, named NIKHITHA V SHETTY, hereby declare that the major project titled **"Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of Master of Computer Applications during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

NIKHITHA V SHETTY	20242MCA0137	Nikhitha

PLACE: BENGALURU

DATE: 08/May/2026.

Abstract

Water-borne diseases pose a major threat to public health, especially in rural and underdeveloped regions where access to clean water and timely monitoring is limited. This project presents a **Smart Advisory Health Monitoring System** that uses machine learning to predict the risk of water-borne diseases and generate early warnings.

The system collects environmental and health-related parameters such as pH level, turbidity, rainfall, temperature, and reported cases of diarrhea and vomiting. These inputs are processed using an **XGBoost machine learning model** to classify the risk level into LOW, MEDIUM, or HIGH. Based on the prediction, the system generates alerts and provides recommendations to prevent disease outbreaks.

The application is developed using a **React-based frontend** for user interaction and a **Flask backend** for processing and API handling. All data is stored in an **SQLite database**, and an email notification system is integrated to send alerts in critical situations. The system also provides visualization features such as charts and history tracking for better analysis.

The system is designed using a client–server architecture, where the frontend interface allows users to input and visualize data, and the backend handles data processing and model prediction. By integrating multi-source data, the system improves prediction accuracy and supports proactive decision-making.

The proposed system enables **real-time monitoring, early detection, and preventive action**, making it an effective solution for improving public health management. It is cost-effective, scalable, and suitable for deployment in resource-limited environments.